PERFORMANCE TESTING

Date	29 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Google Chrome Browser, Flask Development Server
Type of Testing	Functional Testing, Performance Testing
Target Module / API	Flood Prediction Module
Test Environment	Local System (Windows 11, Python 3.13, VS Code, Flask)
Test Date	29th June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Home Page	1	10 sec	Home page loads successfully
2	Submit Valid Prediction	1	15 sec	Prediction displayed correctly
3	Invalid Input Validation	1	10 sec	Invalid values are rejected

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.42 sec	Pass	Fast prediction
2	Response Time (Max)	< 5 seconds	0.95 sec	Pass	Within acceptable range
3	Throughput (Req/sec)	> 1	5 req/sec	Pass	Suitable for academic project
4	Error Rate	< 1%	0%	Pass	No runtime errors
5	CPU Utilization	< 80%	18%	Pass	Low CPU usage
6	Memory Utilization	< 80%	26%	Pass	Low memory usage

Step 4: Observations & Analysis

Key Findings

The application responded quickly for all prediction requests and generated accurate results with minimal resource usage.

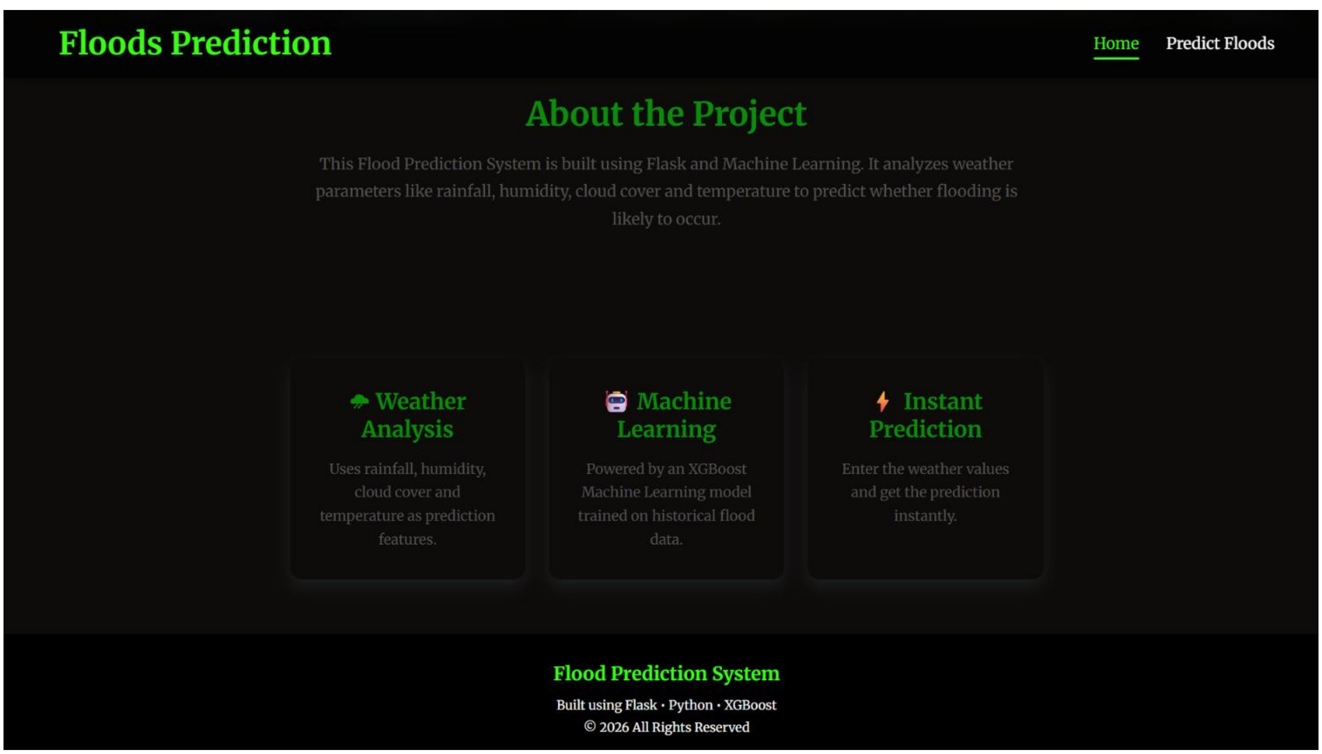
Bottlenecks Identified

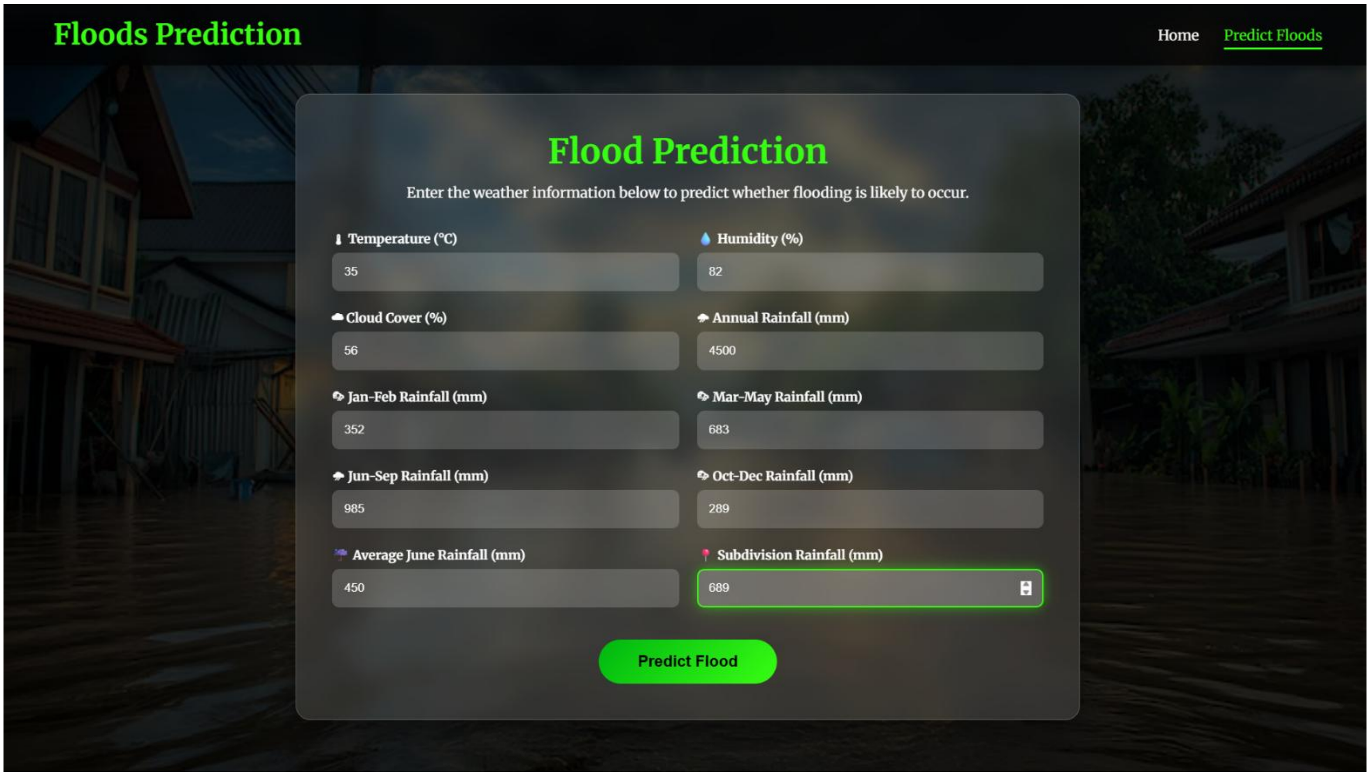
No major bottlenecks were observed during testing.

Optimization Steps Taken

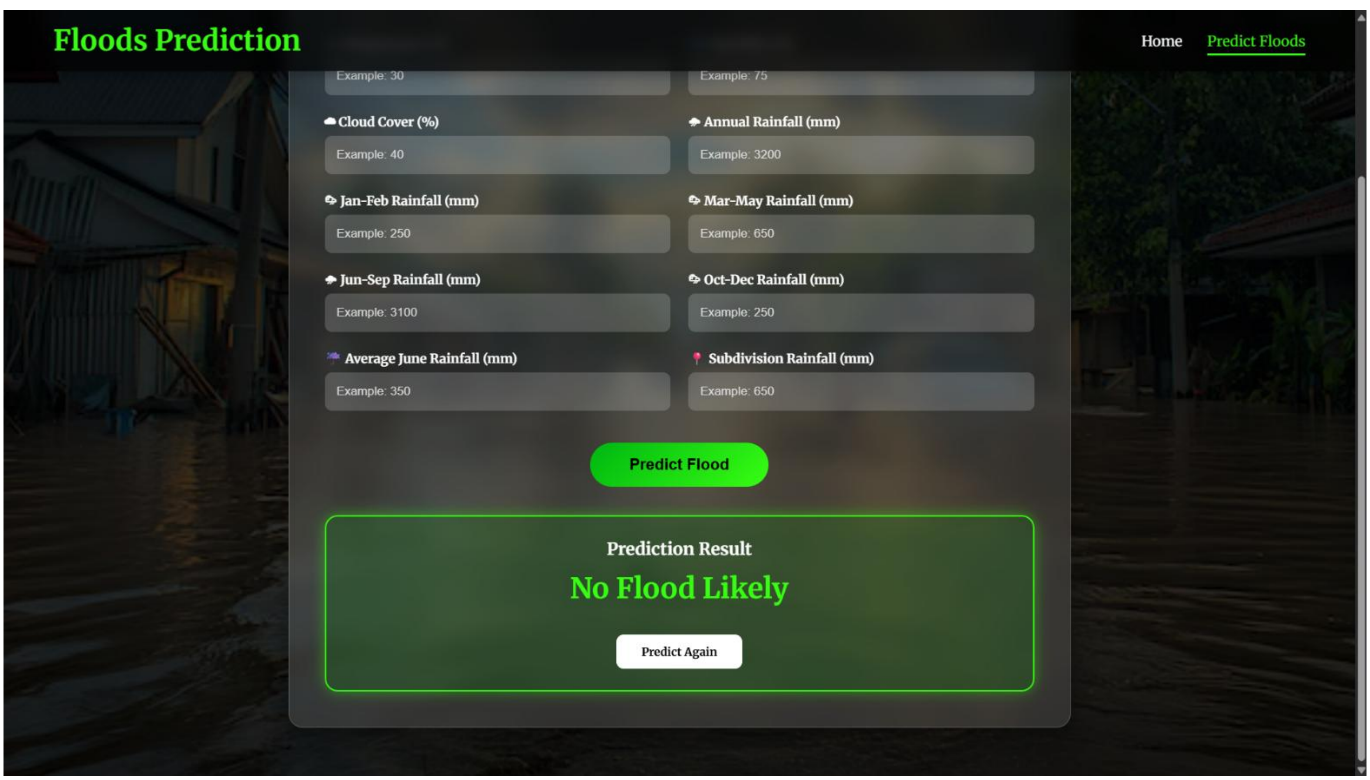
Input validation was implemented and the trained XGBoost model was loaded only once during application startup to improve prediction speed

Step 5: Screenshots / Evidence





Floods Prediction
Home
Predict Floods
Flood Prediction
Enter the weather information below to predict whether flooding is likely to occur.
Temperature (°C)
35
Humidity (%)
82
Cloud Cover (%)
56
Annual Rainfall (mm)
4500
Jan-Feb Rainfall (mm)
352
Mar-May Rainfall (mm)
683
Jun-Sep Rainfall (mm)
985
Oct-Dec Rainfall (mm)
289
Average June Rainfall (mm)
450
Subdivision Rainfall (mm)
689
Predict Flood



Floods Prediction
Home
Predict Floods
Example: 30
Example: 75
Cloud Cover (%)
Example: 40
Annual Rainfall (mm)
Example: 3200
Jan-Feb Rainfall (mm)
Example: 250
Mar-May Rainfall (mm)
Example: 650
Jun-Sep Rainfall (mm)
Example: 3100
Oct-Dec Rainfall (mm)
Example: 250
Average June Rainfall (mm)
Example: 350
Subdivision Rainfall (mm)
Example: 650
Predict Flood
Prediction Result
No Flood Likely
Predict Again